

Neurocognitive Functioning in Tourette Syndrome

From Research Findings to Clinical Implementation



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KEYWORDS

• Tourette syndrome • Tics • Neuropsychology • Cognition

KEY POINTS

- Cognitive assessment in Tourette syndrome may inform implementation of supportive workplace or education accommodations.
- TS often comes with co-occurring conditions, which complexify the literature on cognition.
- Existing literature suggests potential executive functions difficulties, although these may be influenced by co-occurring conditions.
- For most cognitive functions, co-occurring conditions such as attention deficit hyperactivity disorder and obsessive-compulsive disorder are associated with poorer cognitive outcomes.

INTRODUCTION

Neurocognitive functioning is an important area of research in Tourette Syndrome (TS). Several studies have tried to identify cognitive strengths and weaknesses among individuals living with TS. Investigating the cognitive profile of people with TS serves important purposes. It could lead to better targeted treatments, and to classroom and workplace adaptations that would benefit people living with TS.

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Several factors complicate neurocognitive research in TS. Co-occurring conditions are frequent in TS—up to 85% of individuals with TS have at least 1 co-occurring condition.¹ For instance, ~50% of children with TS also have attention deficit hyperactivity disorder (ADHD),¹ which also brings a diverse array of specific cognitive difficulties. Furthermore, subclinical features of neurodevelopmental conditions may also be experienced by a larger proportion of individuals. Other factors, such as medication intake and duration of symptoms, may also confound findings. This review aims to offer an overview of challenges and strengths of individuals with TS across cognitive domains while considering potential confounding factors.

COGNITIVE DOMAINS

Intellectual Functions

Intellectual functioning in TS has mostly been assessed in children and adolescents. Most evidence suggests no major difference in intelligence quotient (IQ) between youth with TS and typically developing youth.^{2–4} Although Debes and colleagues⁵ found reduced performance in the Block Design subtest of the Wechsler Intelligence Scale for Children in children with TS-only, most studies suggest that differences in IQ are attributable to co-occurring disorders (mainly ADHD).^{6,7}

Attention

Given the frequent co-occurrence of TS and ADHD, attention is an important domain to assess in individuals with TS. However, this frequent co-occurrence raises significant conceptual issues. Consideration needs to be given to whether attentional difficulties are likely to meet criteria for a diagnosis of ADHD, or whether sub-threshold attentional difficulties are part of the TS spectrum in those without a co-occurring ADHD diagnosis. For instance, several studies have reported more inattention and hyperactivity symptoms in children with TS without clinically significant ADHD relative to typically developing children.^{6,8,9} It is thus important to keep in mind that ADHD symptoms most likely exist on a continuum among youth with TS.

Sustained attention

Sustained attention reflects the ability to maintain attentional focus over time. Among children, difficulties with sustained attention have mostly been reported in those with co-occurring ADHD^{9–12} or obsessive-compulsive disorder (OCD),^{13,14} while those without co-occurring conditions do not usually differ from typically developing children.^{9,10,12,15} Sustained attention has been assessed less frequently among adults, but available studies mostly reported no differences between those with TS and controls.^{16–18}

Selective attention

Selective attention refers to the ability to focus attention on relevant information while ignoring distractions. Few studies have assessed this function in TS. In the Trail Making Test Part A, children with TS were significantly slower than typically developing controls.¹⁹ Those differences were however accounted for by co-occurring disorders. In a letter cancellation task, no significant difference was found between children with TS and controls.²⁰ In adults, difficulties with selective attention in letter cancellation tasks appear to be mainly present for harder versions of the task²¹ or when co-occurring disorders are present.¹⁸

Divided attention

Divided attention can be understood as the ability to attend to 2 tasks at the same time. In children with TS, it has mostly been assessed with dual-tasks, and difficulties

with such tasks appear to be mainly influenced by co-occurring ADHD and not having TS *per se*.^{10,22,23} In adults, a similar profile emerged,¹⁸ although difficulties with the detection of auditory stimuli were reported when those were presented concurrently with visual targets that were difficult to identify.²⁴

Memory and Learning

Memory is a fundamental process that allows individuals to learn, acquire, retain, and recall information based on interactions with their environment. Memory is not a unitary phenomenon: one fundamental dissociation is between memories available to conscious awareness (explicit memory) and learning features of the environment affecting behavior without conscious awareness (implicit memory).

There are few reports of difficulties with declarative memory amongst children with TS. Prior research showed no difficulties with word list²² or story²⁵ recall. However, children with TS were found to have less specific autobiographic memories than typically developing children.³ Research on explicit memory is also scarce in adults and conflicting findings have been reported, such as intact word list²⁶ but not story²⁷ recall.

There is a growing body of research investigating non-declarative, implicit memory amongst individuals with TS. Examples of implicit memory include the learning/acquisition of non-verbal information, such as procedural memory, sequence learning, priming, and associative learning. Studies on procedural memory in TS have produced mixed findings. Several using serial reaction time tasks have shown intact^{25,28,29} or enhanced³⁰ sequence learning abilities amongst children with TS, as well as long-term consolidation of procedural memories.³¹ However, other studies in children and adults with TS have demonstrated difficulties on probabilistic learning tasks, in which participants must learn associations between stimuli and outcomes, with more difficulties amongst those with greater tic severity.^{32,33} Further investigation of those discrepancies revealed enhanced learning of probabilistic information but difficulties in the acquisition of order-based information.³⁴ While there is a burgeoning research literature on the role of implicit memory in TS, with significant implications in its role in other domains (eg, adaptive behavior), this remains an area that is rarely given consideration in clinical assessment. Interestingly, procedural learning has been implicated as a strong underpinning factor in grammatical processing,³⁵ and it is notable that a number of studies have demonstrated possible enhanced language abilities in the TS population (see as follows).

Language

In a recent review of language and communication in children with TS, Feehan and Charest³⁶ noted that language disorders were more prevalent in children with TS than in the general population, and even more in those with co-occurring conditions. They also highlighted preliminary evidence for challenges with high-level language. Verbal IQ, verbal fluency, and receptive vocabulary skills were mostly within the average range, although co-occurring ADHD or OCD could lead to impairment. The authors suggested that vocabulary, categorization, and the ability to recall or repeat language could be considered as strengths in children with TS. Other areas of strength among youth with TS include speeded morphologic³⁷ and phonologic³⁸ processing.

Fewer studies have assessed language in adults with TS. Most studies report no differences with verbal fluency.^{18,39–41} Difficulties in pragmatic language have been reported in both children⁴² and adults.⁴³

Motor and Visuomotor Skills

Given the involvement of the motor system in TS, motor skills have been extensively assessed in this population, with tasks such as the Purdue Pegboard or the finger

tapping test. In a literature review, Kalsi and colleagues⁴⁴ noted several limitations among this literature, such as small samples and not accounting for co-occurring conditions or medication intake. More recent studies in children with TS have reported difficulties during the finger tapping test⁴⁵ but no difference with controls in the Purdue Pegboard.⁴⁶ In adults, Abramovitch and colleagues⁴⁷ reported a Purdue Pegboard performance that was one standard deviation below the normative data. Additionally, data from children⁴⁸ and adults⁴⁹ with TS performing finger sequence tasks revealed abnormal lateralization of fine motor skills, suggesting that those difficulties may persist across development. Thus, difficulties with motor skills may exist in TS, but more data are needed to reach definite conclusions. Nevertheless, motor skills may have significant importance in TS, as they may be associated with tic severity across development.⁵⁰

Executive Functions

Inhibitory control

Inhibitory control has been extensively assessed in TS, but findings have been mixed. This may be, at least partly, attributed to different and inconsistent experimental approaches, or to a lack of consideration for confounding factors. A meta-analysis of 61 studies revealed that both children and adults with TS showed reduced inhibitory control abilities relative to controls.⁵¹ Whereas slight differences were found between controls and individuals with TS without co-occurring conditions, unsurprisingly, those with co-occurring ADHD were found to have more inhibitory difficulties. Differences were also found across tasks: poorer performance was found in sentence completion tasks, Stroop tasks, circle tracing tasks, and the Continuous Performance Test, but no differences were found for Go/NoGo, stop-signal, and stimulus-response compatibility tasks.

Despite the insight provided by this meta-analysis, it lacked the ability to parse out distinct aspects of inhibitory control. A number of studies have undertaken this endeavor with various experimental paradigms, with authors suggesting that individuals with TS show intact inhibition under volitional control,^{52,53} but aspects of automatic inhibition (ie, when operating with very brief delays) were impaired relative to those without TS.⁵² Also, volitional inhibition can be further parsed into proactive and reactive inhibition. A recent study reported impaired reactive inhibition but intact proactive inhibition in adults with TS, suggesting that they do not differ from controls in their ability to postpone actions, but show reduced deceleration of an already initiated response.⁵⁴

Cognitive flexibility

In TS, cognitive flexibility has mainly been assessed with the Wisconsin Card Sorting Test (WCST) and the Part B of the Trail Making Test (TMT). A meta-analysis of WCST studies revealed that differences between individuals with TS and controls were identified mainly in children.⁵⁵ Children with TS completed fewer categories and showed more perseverations, suggesting more difficulties to adapt their responses to changing instructions, compared with control participants. Similarly, several studies reported reduced cognitive flexibility in TMT Part B^{19,56,57} or in the intra-extra dimensional set-shifting subtest of the Cambridge Neuropsychological Test Automated Battery. Although some studies reported null findings,^{7,25} most evidence points toward difficulties with the cognitive functions needed to adapt one's behavior to rule changes. However, since cognitive flexibility involves other cognitive processes such as inhibitory control, observed differences could be caused by a generalized multi-domain performance deficit or executive function deficit.

In adults, results have been mixed. The aforementioned meta-analysis revealed no difference in terms of WCST categories completed, but slightly more perseverations in participants with TS than controls.⁵⁵ Studies using the TMT Part B found both average range²⁶ and impaired⁵⁸ performances. In other tasks involving switching abilities, such as a number ordering task⁴¹ or an attentional set-shifting test,⁵⁹ adults with TS underperformed compared with controls. Although difficulties in cognitive flexibility may persist in adulthood, those seem less prevalent than in childhood.

Planning

In children with TS, planning skills were mostly assessed with tower tasks, such as the Tower of London. Most studies reported intact planning skills in children with TS,^{60–62} although a few studies unsurprisingly found impaired performance in those with co-occurring ADHD.^{61,63} Similarly, poorer performance in the Six Elements Test was only found for children with TS + ADHD.²⁵ Some studies have also assessed planning with the organization score of the Rey-Osterrieth Complex Figure but reported no deficits in children with TS.^{25,64} Thus, difficulties with planning appear to be limited to children with co-occurring ADHD, although difficulties during the Tower of London has also been reported in children with TS without co-occurring disorders.⁴⁵ In adults, studies have mostly found intact planning skills during the aforementioned tasks.^{26,40,41,65}

Decision-making

Only few studies have assessed decision-making in TS. Among children with TS, a pattern of safer choices in the Iowa Gambling Task was observed.⁶⁶ Among adults, no difference was found between TS and control groups during the Iowa Gambling Task.^{67,68} Similarly, adults with TS were not found to differ from controls regarding decision-making under risk and ambiguity situations (although those with TS had somewhat less consistent decisions).⁶⁹ However, co-occurring ADHD and OCD may lead to a different pattern of risk-taking.

Few studies have assessed temporal discounting in TS, and so far, conflicting findings have emerged. Whereas Vicario and colleagues⁷⁰ found steeper discounting in adolescents with TS, Schüller and colleagues⁷¹ found the opposite in adolescents and no difference in adults. Those differences might be due to different task demands, and thus more research is needed on this topic.

Working memory

Across most tasks assessing working memory, such as the digit span,^{7,22,40,59,72} spatial span,⁷³ Corsi span,⁶² n-back,^{67,74} digit ordering,^{39,75,76} self-ordered pointing,⁴⁰ and letter-number sequencing tests,⁷⁷ children and adults with TS tend to perform to a comparable level to typically developing controls (although see^{58,78,79}). However, factors such as the presence of co-occurring ADHD^{61,63,80} or OCD,¹⁸ as well as increased tic severity⁸¹ may lead to more difficulties in working memory.

Subjective ratings of everyday executive function

Informant- and self-rated measures of executive function, such as the Behavioral Rating Inventory of Executive Function (BRIEF) or the Childhood Executive Functioning Inventory (CHEXI), provide an important counterpart to experimental and objective measures, as they have the advantage of describing an individual's day-to-day functioning. They may also assess distinct (and potentially complementary) aspects of executive functions. For instance, one study reported that children with TS did not differ from typically developing controls on a lab-based measure of inhibitory control (ie, stop-signal reaction time) but had more impaired scores on the BRIEF Inhibit subscale.⁸²

Several other studies have also reported challenges in everyday executive functions.^{79,83–88} ADHD may be the driving factor behind those differences. Studies have reported diminished subjective executive function in children with TS + ADHD compared with typically developing children,^{79,82,87} and for some domains, with children with TS-only.^{79,82,86} There may also be developmental aspects to consider, as differences between TS and TS + ADHD have been reported for children up to 12 years old but not older adolescents.⁸⁶ It is not clear whether TS, independently of ADHD, results in poorer everyday executive functions. On the one hand, some studies have found differences in specific BRIEF indices between children with TS without co-occurring ADHD and typically developing controls.^{82,87,88} On the other hand, Hamilton and Harrison⁸³ reported that differences between children with TS and controls regarding CHEXI measures, even in the absence of a formal ADHD diagnosis, were driven by ADHD-like characteristics.

Creativity and Problem-solving

A few studies have assessed problem-solving abilities in TS with design fluency tasks. Overall, preserved design fluency was found in both children^{15,89} and adults;¹⁸ however children with TS use less numerical strategies than typically developing controls.⁸⁹ A recent study assessed divergent and convergent thinking, which respectively refer to the ability to generate several original solutions to a problem and to find a unique solution to a problem by making links with concepts already known.⁹⁰ Adults with TS were better than controls at convergent thinking, but groups did not differ in terms of divergent thinking. However, increased divergent thinking skills was found to reduce subjective tic-related discomfort.

Social Cognition

The study of social cognition in children with TS is relatively recent. Difficulties with sarcasm comprehension have been reported in children with TS-only and TS + ADHD.⁴² Children with TS + ADHD were also found to have auditory emotional recognition difficulties, especially for anger.⁷⁷ Adolescents with TS may also endorse more utilitarian solutions than controls when presented with moral dilemmas.⁹¹

In a literature review published in 2013, Eddy and Cavanna⁹² highlighted differences pertaining to theory of mind, distinguishing self-other mental states, nonliteral language, and sarcasm comprehension, as well as difficulties with social reasoning and social decision-making. More recent studies also reported difficulties with recognizing angry prosody⁷⁷ and visual emotion recognition,⁹³ as well as increased hypermentalizing⁵⁸ and more frequent use of emotional suppression strategies for negative emotions.⁹⁴ Thus, while the study of social cognition is fairly recent in TS, there appears to be the suggestion of some difficulties.

Adaptive Behavior

Adaptive behavior is rarely measured in studies but is a core aspect of assessment, given the increased risk of impairment in everyday function associated with challenges in some cognitive domains. Studies have reported adaptive difficulties in children with TS and co-occurring disorders (mostly ADHD) on the Socialization,^{95–99} Daily living,^{95,99} and Communication^{95,99} scales of the Vineland Adaptive Behavior Scales. Difficulties with socialization may also be present even when controlling for co-occurring ADHD.⁹⁶ A similar picture emerged for the competence scales of the Child Behavior Checklist, where children with TS and co-occurring disorders scored below normative data or controls.^{95,96,98} Those weaknesses may remain even when controlling for co-occurring ADHD.^{95,96}

Table 1
Main findings across cognitive domains

Domain	Main Findings	Developmental Perspective	Impact of Co-occurring Disorders
Intellectual functions	Mostly intact in TS.	Mostly assessed in youth, not much is known in adults.	Potentially impacted by co-occurring ADHD.
Attention	Most differences attributable to co-occurring ADHD or subthreshold ADHD symptoms.	Insufficient data to draw conclusions.	Mostly impacted by co-occurring ADHD.
Memory and learning	Inconsistent findings: reduced and intact performances have been reported, with strengths in certain aspects of implicit memory.	Insufficient data to draw conclusions.	Insufficient data to draw conclusions.
Language	Some difficulties, such as pragmatic language. However, also strengths in specific aspects of language.	Insufficient data to draw conclusions.	Lower in individuals with co-occurring disorders (ADHD, potentially OCD as well).
Motor function and spatial cognition	Inconsistent findings, but potential difficulties.	May predict adult tic severity outcomes.	Insufficient data to draw conclusions.
Executive functions	Inhibitory control, cognitive flexibility and every day EF are slightly reduced. Planning and working memory are mostly intact whereas decision-making has inconsistent findings.	No difference between youth and adults in inhibitory control. More difficulties with cognitive flexibility in children, relative to adults. Insufficient data to draw conclusions in other domains.	Poorer or potential for poorer function in context of co-occurring ADHD.
Creativity and problem solving	Mostly intact in TS. Potential strength in creativity.	Insufficient data to draw conclusions.	Insufficient data to draw conclusions.
Social cognition	Difficulties in several aspects, including nonliteral language comprehension.	Few studies in children so far.	Potential impact of co-occurring disorders, not well-assessed throughout the literature.

Note: ADHD, attention deficit hyperactivity disorder; EF, executive functions; OCD, obsessive-compulsive disorder; TS, Tourette syndrome.

DISCUSSION

In the previous sections, we highlighted differences in cognitive functioning between individuals with TS and controls. These differences not only suggest difficulties but also strengths in various areas. However, the literature, to this date, does not provide a clear neuropsychologic profile of individuals with TS. Despite this lack of a clear neurocognitive endophenotype, cognitive assessment is an important part of TS assessment, where impact of challenges is seen in an individual's daily life. Below, we highlight areas where cognitive assessment and research on neurocognition may benefit individuals living with TS.

Impacts on Daily Functioning

Cognitive assessment is important because lower educational and occupational achievements are reported in studies of individuals with TS and may be the consequence of neuropsychologic sequelae.^{100–103} The direct impact of cognitive difficulties on educational outcomes has scarcely been investigated, but IQ and attention were found to predict arithmetical achievement in children with TS.¹⁰⁴ Also, poorer parent-rated everyday executive functions have been associated with increased sensory dysregulation,⁸⁸ as well as more impaired daily life at home and poor expectations for the future.¹⁰⁵

Cognitive Functions and Treatment

Assessing cognitive functions in TS is important as they could potentially change during treatment, predict treatment outcome, or indicate which treatment might be best suited for someone. For instance, inhibitory control has often been hypothesized to play a role in behavior therapy for tics. However, few studies have explored whether it predicts treatment outcome. Two studies have found a potential role of inhibitory control as a nonspecific predictor of treatment outcome.^{106,107} However, in analyses where behavior therapy was contrasted with a control treatment^{47,108} or with treatment-as-usual,¹⁰⁹ inhibitory control did not moderate treatment outcome. It is also unclear whether behavior therapy improves inhibitory control abilities. While Petruo and colleagues¹¹⁰ found improvements in inhibitory control in a small sample of adults who underwent behavior therapy, 2 larger studies reported no impact of behavior therapy on inhibitory control abilities.^{107,108}

Motor skills may also be involved in behavior therapy for tics. Following cognitive-psychophysiological treatment, improved motor skills¹¹¹ and changes in movement-related neural activity^{111,112} were found. However, those studies were all conducted in adults. It remains unknown whether those findings would generalize to children.

SUMMARY

This article reviewed the main strengths and difficulties across several cognitive domains (summarized in [Table 1](#)) in youth and adults with TS. As mentioned, those strengths and difficulties are not only influenced by the presence of tics but also by other factors such as the presence of co-occurring disorders and medication intake. A longer clinical history of tics has also been associated with negative impacts on cognitive functioning.² However, as longitudinal research is rare in TS, these findings are yet to be replicated robustly. We therefore recommend future studies to document and control those factors so that findings can be better contextualized and more readily transferable to decision-making and advice for patients within clinical practice.

CLINICS CARE POINTS

- Balance of hypothesis-driven assessment based on referral question, as well as information from the individual and informants, alongside being informed by the literature.
- Key elements of formulation to guide evidence-based intervention and understanding of the individual's needs, given due consideration to subthreshold symptoms for ADHD and OCD.
- Triangulation of performance of standardized measures, self/informant report, and available resources for remediation, accommodation, adaptations, and self-advocacy.
- Importance of ensuring that everyone (referrers, individual with TS and their families, teachers, and employers) understand the findings (both strengths and weaknesses) of the assessment and can make use of the material.

DISCLOSURE

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